

Training-Free Token Reduction on MotionBench

Results

VLLM Project

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Eleven training-free video-LLM token-reduction methods, evaluated on MotionBench (8,052 questions, 4,018 scoreable) across two backbones — LLaVA-OV-7B and Qwen3-VL-8B — at 32 frames. Results only: full methodology, the verification gate, and limitations are in `paper.tex`. At $n = 4,018$ the binomial 95% confidence half-width is ± 1.54 points; * marks a difference significant by McNemar’s test on paired predictions ($\chi^2 \geq 3.84$, $p < 0.05$).

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1 Common settings

Held fixed across every method and both backbones:

- **32 frames**, sampled uniformly per video.
- **Greedy decoding** (`do_sample=False`, `max_new_tokens=16`), batch size 1.
- **Letter-match scoring** (A–D) against ground truth; the 4,034 NA items are excluded, leaving 4,018 scoreable.
- Methods exposing a single retention knob are standardized to **15% nominal retention**; methods with no such knob (frame-selection methods, or methods whose budget is fixed internally) keep their published defaults — forcing 15% on those would break paper fidelity.

Per-method settings, mechanisms and measured token budgets are in Table 6 (§4).

2 Master table

Table 1 and Table 2: every method at its standardized 15% or published default, one table per backbone; footnotes mark exceptions (frame-selection methods, no-knob methods, published alternate points). Full retention sweeps are in §2.1.

Table 1: LLaVA-OV-7B, 32 frames, 15% nominal retention except where noted. AO = Action Order, CM = Camera Motion, LM = Location-related Motion, MR = Motion Recognition, MO = Motion-related Objects, RC = Repetition Count (% correct within category, chance = 25%). Δ is vs. baseline. * = significant. § AIM: 25% retention (two active merge steps ship in the released code). † PruneVID-OV: `cluster_ratio=0.5` (50%), the authors’ only published point. ‡ no retention knob; one fixed recipe (Table 6). ¶ VisionZip is the contextual half only — no CLS token to select the dominant half (same as the Qwen3-VL cell, Table 2).

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
<i>Baseline</i>	<i>52.66</i>	—	<i>40.5</i>	<i>45.2</i>	<i>55.5</i>	<i>57.0</i>	<i>71.2</i>	<i>23.8</i>
DyCoke [‡]	53.36	+0.70	38.9	48.8	55.7	58.2	70.9	25.2
FlashVID	53.31	+0.65	39.9	45.7	53.8	58.0	71.7	28.2
HoliTom	53.14	+0.47	40.8	49.6	52.6	57.0	71.4	27.3
MDP3	53.06	+0.40	40.5	49.6	53.5	56.8	71.6	26.5
VideoITG	52.86	+0.20	40.1	47.0	53.7	57.6	70.1	26.8
AIM [§]	52.86	+0.20	41.4	48.1	54.2	57.0	71.9	22.2
VisionZip (ctx.) [¶]	51.79	−0.87	40.1	46.2	53.7	56.2	69.4	23.2
STTM [‡]	51.72	−0.95	39.9	48.1	53.8	53.5	70.6	28.7
PruneVID-OV ^{*†}	38.20	−14.46	32.0	34.8	35.5	38.4	52.9	27.3
FastV [*]	36.78	−15.88	32.8	31.9	34.1	35.7	53.8	25.2

Table 2: Qwen3-VL-8B, 32 frames, 15% nominal retention except where noted. All ten portable methods. VisionZip is the contextual half only, as in Table 1. FlashVID and STTM run the authors’ own code at published settings (§4). † PruneVID: `cluster_ratio=0.5` (50%), the authors’ only published point. ‡ no retention knob; one fixed recipe (Table 6).

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
<i>Baseline</i>	<i>62.52</i>	—	<i>46.1</i>	<i>63.1</i>	<i>65.0</i>	<i>67.3</i>	<i>79.0</i>	<i>33.8</i>
PruneVID [†]	62.17	−0.35	46.1	62.9	64.1	67.1	79.1	32.5
DyCoke [‡]	61.85	−0.67	45.1	62.6	63.9	66.4	78.4	34.5
STTM ^{*‡}	61.60	−0.92	45.9	61.6	64.8	66.0	77.1	34.8
HoliTom [*]	60.33	−2.19	44.7	61.3	61.0	66.4	76.2	29.0
MDP3 [*]	59.66	−2.86	44.9	64.7	62.8	64.0	77.4	23.0
FlashVID [*]	59.36	−3.16	43.5	57.4	62.6	65.6	77.2	23.5
FastV [*]	59.01	−3.51	46.2	61.0	61.9	63.5	72.2	30.5
VisionZip (ctx.) [*]	58.81	−3.71	43.2	57.4	61.9	64.2	74.3	29.5
VideoITG [*]	56.35	−6.17	45.3	53.5	59.3	60.2	75.4	22.2
AIM [*]	55.97	−6.55	45.9	56.6	59.5	58.3	72.2	27.0

2.1 Retention sweep

Three methods (FastV, FlashVID, HoliTom) have a retention knob the authors validate at more than one point; each is run at every point they publish, not just 15%. Table 3 and Table 4 are that data, one table per backbone; baseline is in Table 1/ 2 above, not repeated here.

Table 3: LLaVA-OV-7B retention sweep (baseline 52.66%). * = significant. FastV’s range is its authors’ full published boundary, re-run with the `PrunableDynamicCache` fix (§4) — every point reproduces its pre-fix value almost exactly. FlashVID and HoliTom publish 10/15/20/25% on every backbone they support; VisionZip (contextual-only, §2) was swept over the same range for this report.

Method	10%	15%	20%	25%	50%	75%
FastV	36.06*	36.78*	36.93*	36.73*	36.31*	35.69*
FlashVID	52.51	53.31	53.71*	53.36	—	—
HoliTom	52.76	53.14	53.06	53.41	—	—
VisionZip (ctx.)	51.69	51.79	51.92	52.54	—	—

Table 4: Qwen3-VL-8B retention sweep (baseline 62.52%). * = significant. All four FlashVID cells now use the authors’-code implementation (§4), the same one that produced the 15% cell elsewhere in this report — monotonic in retention. FlashVID and HoliTom are not published beyond 25% on this backbone.

Method	10%	15%	20%	25%	50%	75%
FastV	56.92*	59.01*	59.83*	60.60*	62.32	62.64
FlashVID	58.04*	59.36*	60.05*	60.70*	—	—
HoliTom	60.05*	60.33*	60.38*	60.43*	—	—

FlashVID’s 20% cell is a small local peak on LLaVA-OV (53.71% vs. 53.31%/53.36% at 15%/25%), gate-clean but not persisting outside 20%. On Qwen3-VL it’s instead cleanly monotonic (58.04→60.70% over 10–25%). HoliTom shows no such bump on either backbone. VisionZip’s sweep (51.69–52.54%) is gentle and never significant — unlike FastV and PruneVID-OV, it does not collapse on LLaVA-OV. FastV is the sharpest backbone contrast here: collapsed at 50%/75% on LLaVA-OV (35.7–36.3%) but at baseline parity on Qwen3-VL (62.32%/62.64%, not significant) — the same rule that permanently breaks the model on one backbone is harmless by half retention on the other.

PruneVID publishes one operating point (`cluster_ratio=0.5`, its row in Table 1/2); Table 5 is our own additional sweep, not authors’-validated.

Table 5: PruneVID beyond its one published point (not authors’-validated). Qwen3-VL is monotonic in retention; LLaVA-OV is not.

Backbone	10%	25%	50% (published)
LLaVA-OV-7B (OV port)	38.10*	38.38*	38.20*
Qwen3-VL-8B	60.23*	61.40*	62.17

Tripling FastV’s budget from 10% to 25% moves it 3.68 points on Qwen3-VL, against no more than 0.85 points for any LLaVA-OV method over the same range — the two backbones respond to retention completely differently.

3 Figures

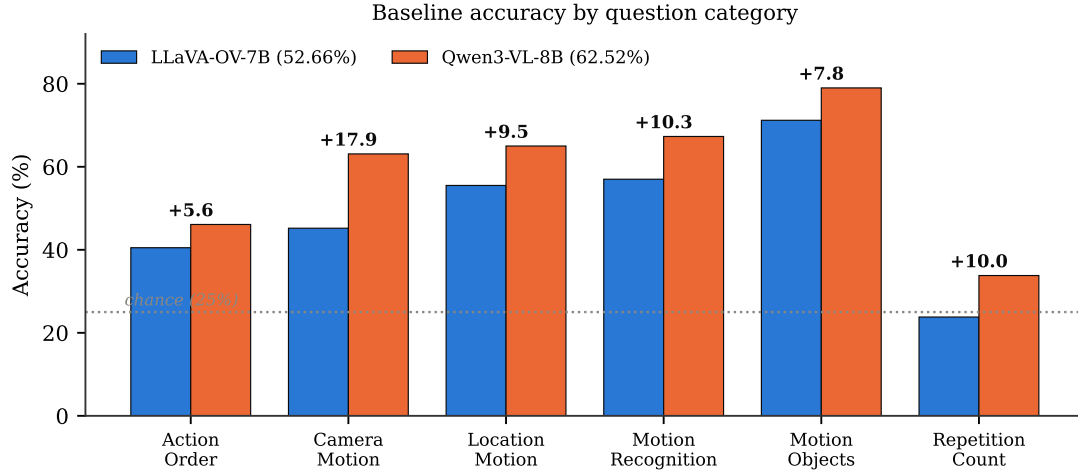


Figure 1: Baseline accuracy by category, both backbones (bar).

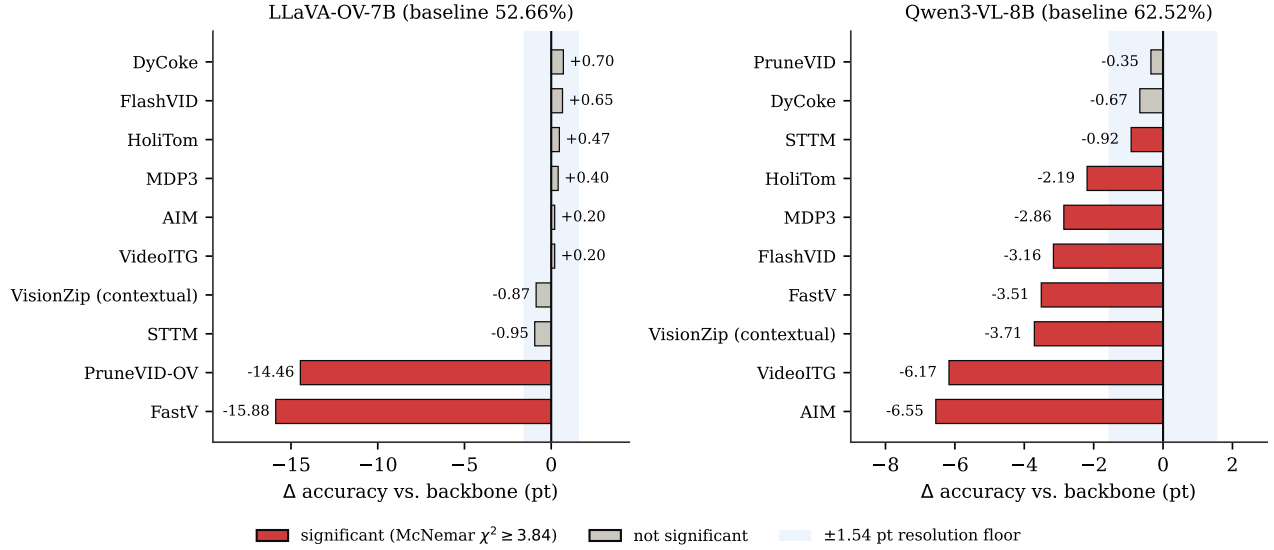


Figure 2: Change vs. backbone, every method, both backbones (bar). Red = significant by McNemar’s test; the shaded band is the ± 1.54 -point resolution floor.

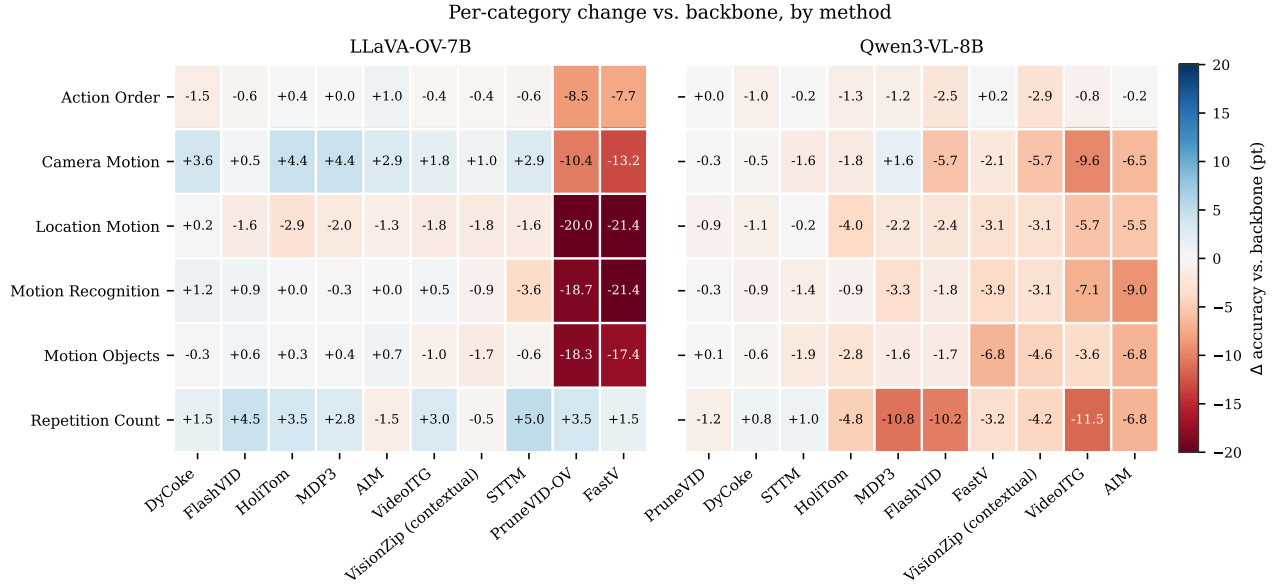


Figure 3: Per-category change vs. backbone, by method (heatmap). Blue = gain, red = loss.

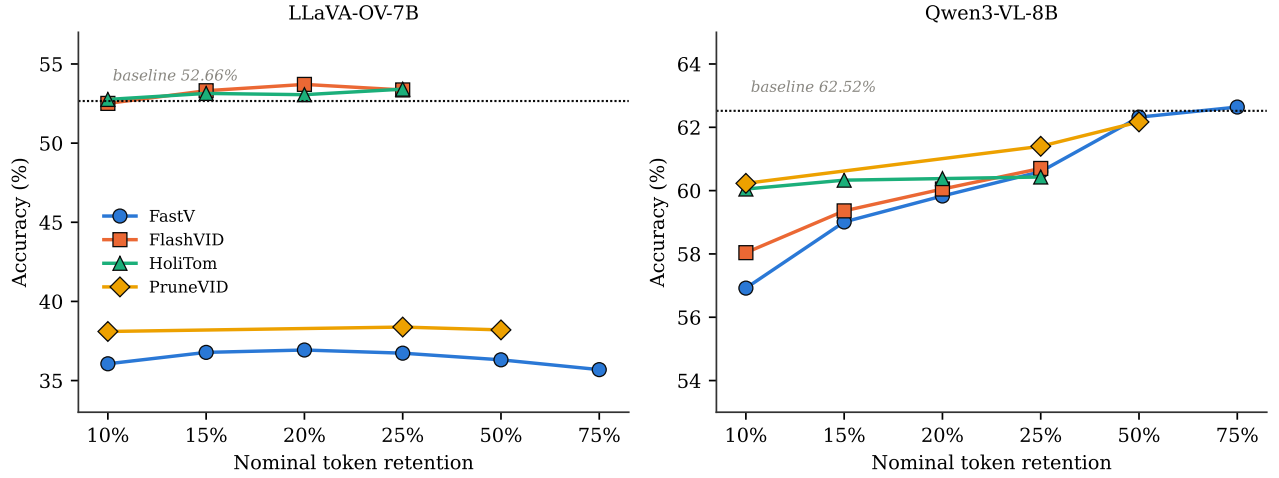


Figure 4: Accuracy against nominal token retention, the four methods with a retention knob (dot + line).

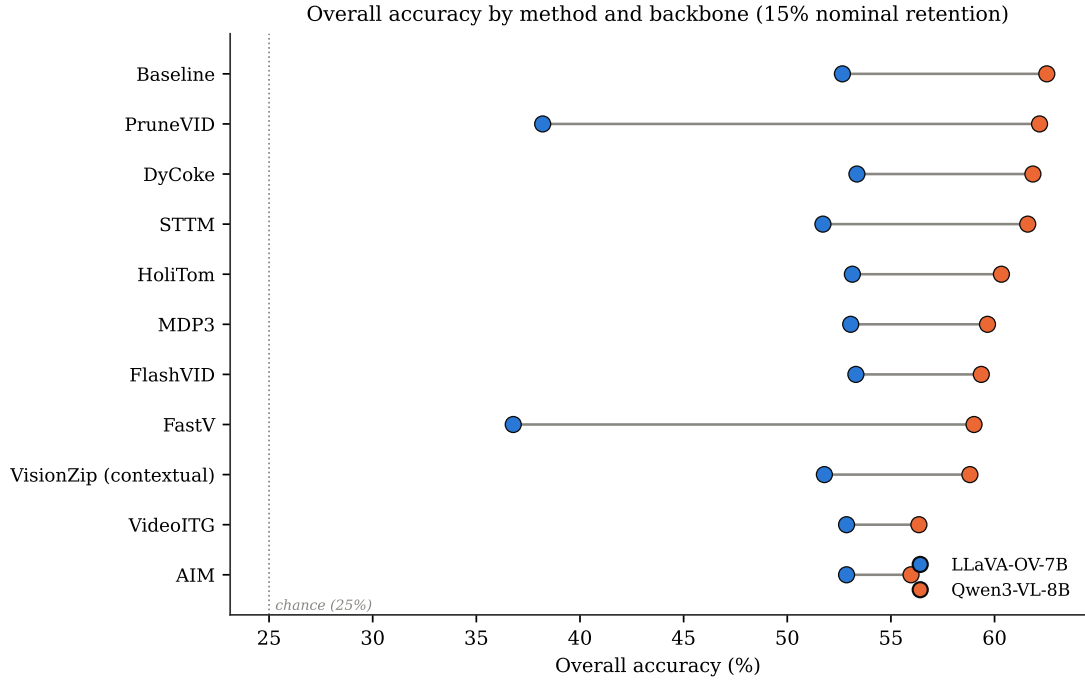


Figure 5: Overall accuracy by method and backbone at 15% nominal retention (Cleveland dot plot). Rows sorted by Qwen3-VL accuracy; the line spans each method’s two backbone results.

4 Per-method settings

Table 6 is the mechanism and exact configuration behind every row above.

Table 6: Per-method mechanism, key parameters (as run in this report), and measured visual-token budget. Budgets are read from each method’s execution log on Qwen3-VL-8B (11,664 visual input tokens at 32 frames), not from its CLI flag — nominal 15% describes the configured knob, not necessarily what is actually retained. Frame-selection methods have no token budget to measure; they instead reduce the number of frames fed to the model.

Method	Mechanism	Key parameters (this report)	Measured budget
HoliTom	pre-LLM segment retain + in-LLM merge	RETAIN_RATIO=0.15 T=0.80 HOLITOM_k=18 HOLITOM_r=0.5	7.5% (875 tok.)
STTM	quadtree spatio-temporal merge	LLaVA-OV: thresh=0.85 temporal=0.65 root=1. Qwen3-VL: thresh=0.85 temporal=0.65 root=2 — a published setting (s3.sttm_pub_run), replacing an earlier off-spec root=0, temporal=-1.0 cell	9.0% (1,050 tok.) on LLaVA-OV; not re-measured at the corrected Qwen3-VL setting
FastV [†]	in-LLM attention rerank, layer 2	K=2, R=0.85 (keep 15%). Re-run with the PrunableDynamicCache fix; K=2/R=50% (36.31%) and K=2/R=75% (35.69%) were also run and land in the same collapse band (Table 3)	15.0% (1,750 tok.)
FlashVID	pre-LLM temporal-segment merge + inner-LLM compression, layer 28	authors’ flashvid() package: retention_ratio=0.15 alpha=0.7 temporal.threshold=0.8 token_selection_method=attn_div pruning_layer=28 llm.retention_ratio=0.1 expansion=1.25 min_segment_num=4	15.0% (1,750 tok.) pre-LLM; the layer-28 stage compresses surviving visual tokens by a further 90% mid-network, not captured by this single figure
AIM	bipartite merge + PageRank prune	2-step merge schedule compiled in; no CLI retention knob. Qwen3-VL 15%; LLaVA-OV 25% (ships two active steps)	15.0% (1,750 tok.)
VisionZip (contextual)	key-similarity merge in the vision tower	LLaVA-1.5-7B (native): dominant=54 contextual=10 at 8 frames. LLaVA-OV & Qwen3-VL: contextual-only port, retention_ratio=0.15 — neither vision tower has a CLS token to select the dominant half	15.0% (1,750 tok.)
DyCoke	temporal merge + KV prune, layer 3	l=3 p=0.7 k=0.7; no single retention knob, net ~49%	49.0% (5,716 tok.)
PruneVID	DPC-KNN cluster merge, layer 10	cluster=0.50 seg=0.25; LLaVA-OV port has no upstream reference (upstream targets PLLaVA only)	50.0% (5,832 tok.)
MDP3	frame selection (DPP)	pool=32 select=8 — selects frames, not tokens	n/a (8 of 32 frames)
VideoITG	grounded frame selection	512 sampled → 32 selected	n/a (32 of 512 sampled)
DyTo [†]	FINCH clustering + ToMe token merge	temporal_aggregation=spatial_tome_finch_dynamic_all_frms; own Vicuna-7B backbone	~3,680 tok. from 100 input frames

[†] DyTo does not run on either standardized backbone; listed here for completeness (see Table 8). [‡] every prior LLaVA-OV FastV run silently discarded its own pruning (a PrunableDynamicCache bug, §7); re-run with the fix, every point reproduces its prefix value almost exactly.

5 Correct/incorrect vs. baseline

Fixed = wrong under the baseline, correct under the method. *Broke* = correct under the baseline, wrong under the method. Both are McNemar discordant pairs on the 4,018 scoreable items. *Differ* counts predictions unequal to the baseline’s over all 8,052 rows (0 would mean a silent no-op — see `paper.tex` for the verification gate this guards against).

Table 7: Paired-prediction statistics against each backbone’s baseline.

Method	Differ	Fixed	Broke	χ^2	Significant
<i>LLaVA-OV-7B</i>					
DyCoke	1,031	170	142	2.34	no
FlashVID	1,610	268	242	1.23	no
HoliTom	1,838	299	280	0.56	no
MDP3	1,452	246	230	0.47	no
VideoITG	1,193	192	184	0.13	no
AIM	1,406	227	219	0.11	no
STTM	4,859	329	367	1.97	no
VisionZip (ctx.)	1,725	260	295	2.08	no
PruneVID-OV	4,536	473	1,054	220.30	yes
FastV	4,701	475	1,113	255.52	yes
<i>Qwen3-VL-8B</i>					
PruneVID	1,074	56	70	1.34	no
DyCoke	1,048	83	110	3.50	no
HoliTom	1,657	131	219	21.63	yes
MDP3	1,626	159	274	30.01	yes
FastV	2,029	149	290	44.65	yes
VisionZip (contextual)	2,035	191	340	41.25	yes
STTM	1,510	133	170	4.28	yes
FlashVID	2,069	157	284	36.00	yes
VideoITG	2,522	206	454	92.44	yes
AIM	2,861	194	457	105.44	yes
<i>Backbone vs. backbone</i>					
Qwen3-VL-8B vs. LLaVA-OV-7B	7,081	810	414	127.47	yes

Divergence volume does not predict damage. STTM on LLaVA-OV changes 4,859 predictions — more than FastV’s 4,701 — yet loses 0.95 points and is not significant, because its changes fall almost evenly in both directions (329 fixed, 367 broken). FastV changes a comparable number destructively (475 fixed, 1,113 broken).

6 Methods on non-standard backbones

Table 8: Runs on backbones other than the two standardized ones. Not comparable to Tables 1–2; no Δ is given because no matched baseline exists for any row. PruneVID’s released code targets PLLaVA only — this is the authors’ real backbone and the one cell with an upstream reference to validate against, unlike the LLaVA-OV port (PruneVID-OV, Table 1).

Method	Backbone	Overall	AO	CM	LM	MR	MO	RC
VisionZip	LLaVA-1.5-7B	40.09	33.7	33.0	37.2	41.1	57.4	25.8
PruneVID	PLLaVA-7B	44.13	35.6	33.0	41.0	47.0	63.3	26.5
DyTo (recon. TW-FINCH)	Vicuna-7B	42.06	32.4	33.0	42.1	43.0	61.7	26.0

7 Notable

The backbone swap moves accuracy 9.86 points — more than any token-reduction method moves it, either direction, either backbone, any retention level tested. Rankings do not transfer: PruneVID is the worst LLaVA-OV result (38.20%) and the best Qwen3-VL one (−0.35); Figure 5 shows the crossing lines directly. FastV and PruneVID-OV collapse specifically on LLaVA-OV (37–38%, > 14 points below baseline, significant), while every other LLaVA-OV method is statistically indistinguishable from the backbone; FastV’s own published range does not save it (35.7–36.3% at 75% and 50% keep, Table 3). Qwen3-VL inverts the pattern: every method loses, eight of ten losses are significant, but none approaches the LLaVA-OV collapse — the largest Qwen3-VL loss (AIM, −6.55) is smaller than the smallest LLaVA-OV collapse.

Only three methods qualify as having an authors’-validated retention sweep (§2.1); the other five with any retention-like parameter do not. **PruneVID** publishes exactly one ratio. **DyCoke** has no single retention knob, only one fixed three-parameter recipe. **AIM** has no knob at all on LLaVA-OV and an unpublished, self-ported one on Qwen3-VL. **STTM** is configured per-benchmark, not a sweepable percentage. **VisionZip**’s contextual-only port does have a sweepable ratio, but the authors never validate one — our 10–25% sweep (Table 3) is ours, not theirs.

A cache bug silently discarded FastV’s and PruneVID’s pruning on LLaVA-OV, not that it mattered. Both prune via a shared `PrunableDynamicCache` class whose `update()` pruned only what it returned to attention, not what it stored — so the pruning was lost every time the cache round-tripped through `generate()`’s legacy-tuple export, and every decode step ran fully unpruned. (PruneVID had a second bug on top: a stray `dycoke=True` left over from a copy-pasted loader, which reset its cache every decode step and crashed once the first bug stopped masking it.) Both are fixed — prune is now physical, survives the round-trip, and the stray flag is removed. Re-running FastV’s full range with the fix reproduces every pre-fix number to within 0.03%: the LLaVA-OV collapse is genuine model behavior, not the bug. PruneVID’s fixed sweep is still running. Qwen3-VL is unaffected (§4: additive-mask implementation, never this cache class).